Predicting Stock Returns Using SPY ETF Data

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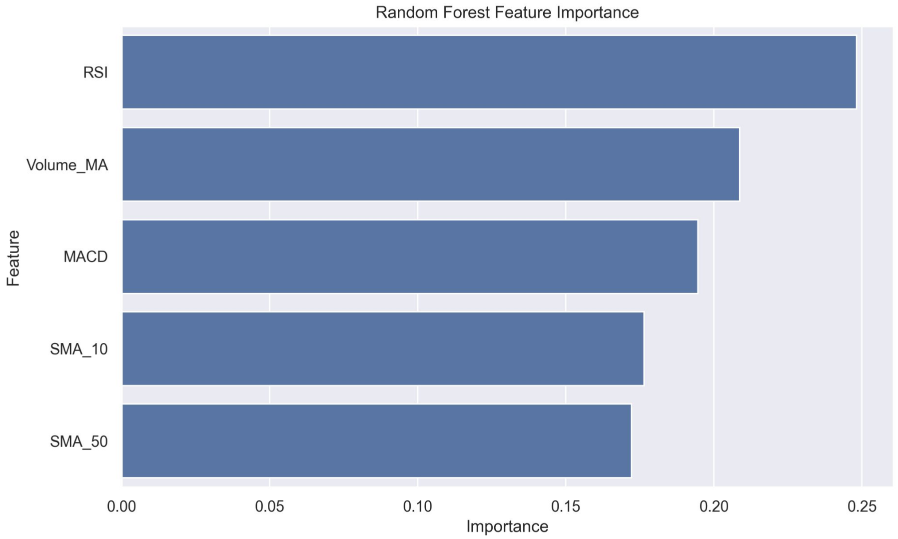
Executive Summary

- This project aims to test whether certain machine learning and regression models can outperform a benchmark buy & hold strategy in predicting daily/weekly stock returns, and potentially identify which features of stocks are most predictive
- Three models were tested: linear regression, logistic regression, and a random forest classifier
- The following features were used as predictors: 10-day Simple Moving Average (SMA), 50-day SMA, Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), Volume Moving Average (VMA)

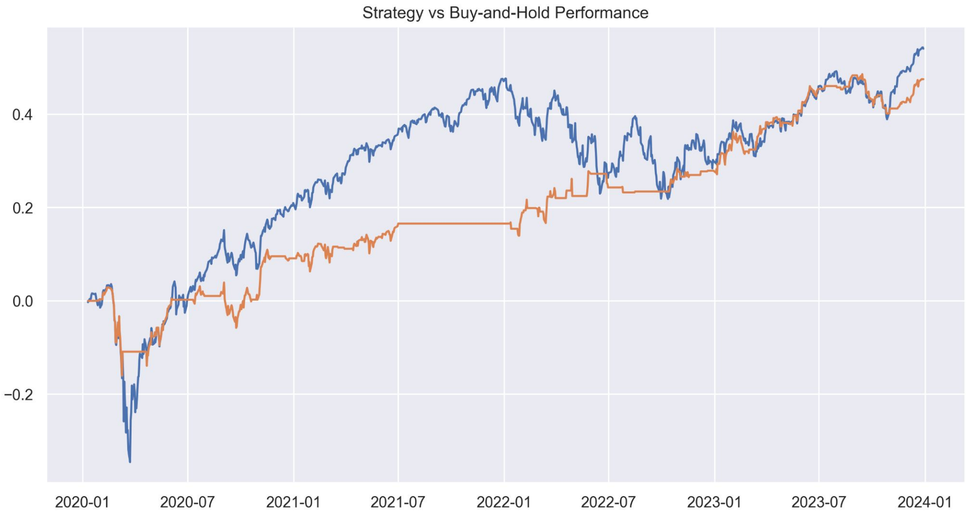
Visualizations



Comparing feature importance of predictors in random forest classifier model:

- RSI is the most predictive feature, with MACD and VMA also being important features, and 10-day SMAs are more important than 50-day SMAs
- RSI and MACD being features with high importance validates mean-reversion strategies and confirms that trend-following signals (MACD line crosses) add predictive power respectively
- VMA being another predictive feature also suggests that liquidity and trading activity are strong signals

Visualizations



Comparing performance of random forest classifier against benchmark strategy:

- Performance is comparable in 2020 and from 2022-2024
- Benchmark significantly outperforms the proposed strategy between 2021 and the first half of 2022
- Indicates that the proposed ML strategy should continued be modified by adding additional features and/or removing existing ones.

Assumptions & Risks

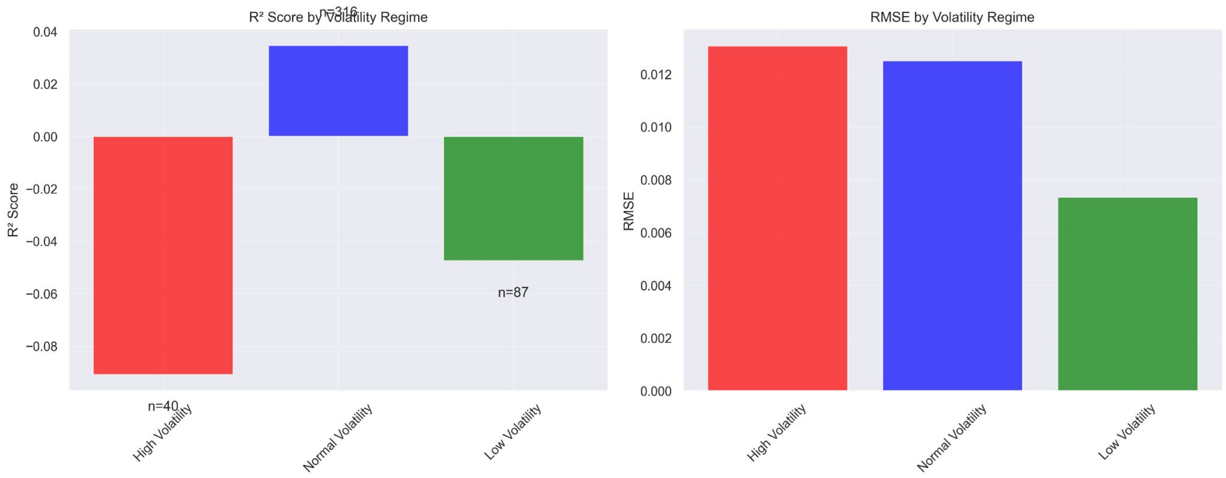
Key Assumptions

- Linearity: Assumes linear relationships between technical indicators and returns
- Independence: Assumes errors are not correlated across time
- Homoscedasticity: Assumes constant error variance across predictions
- Normality: Assumes residuals follow normal distribution
- No Multicollinearity: Assumes features are not highly correlated

Major Risks

- False Confidence: Minimal R^2 could be misinterpreted as predictive power
- Overfitting Risk: Model may capture noise rather than signal
- Regime Dependency: Performance varies significantly across market conditions
- Tail Risk Underestimation: Normal distribution assumptions underestimate extreme events

Sensitivity Analysis



Linear Regression Sensitivity Analysis of Varying Volatility Levels:

- Model performs worst in high-volatility regimes (RSI < 30)
- Better performance in normal conditions (30 <= RSI <= 70), but still minimal predictive power
- Demonstrates the model's limited flexibility in adapting to different market environments

Decision Implication

The proposed ML model should be modified based on the results of model performance assessment, backtesting, and sensitivity analysis

- Low R-squared value, underperformance against benchmark, limited adaptability
- Future work: Expand on existing feature engineering by adding more features or modifying current ones

Other models can be considered for implementing and testing against benchmark:

- Examples: Gradient boosting, neural networks, ARIMA/GARCH models

Consider adding in additional data: sentiment, macroeconomic indicators, order flow